

Synthetic Futures: Decolonising design futuring through synthetic data

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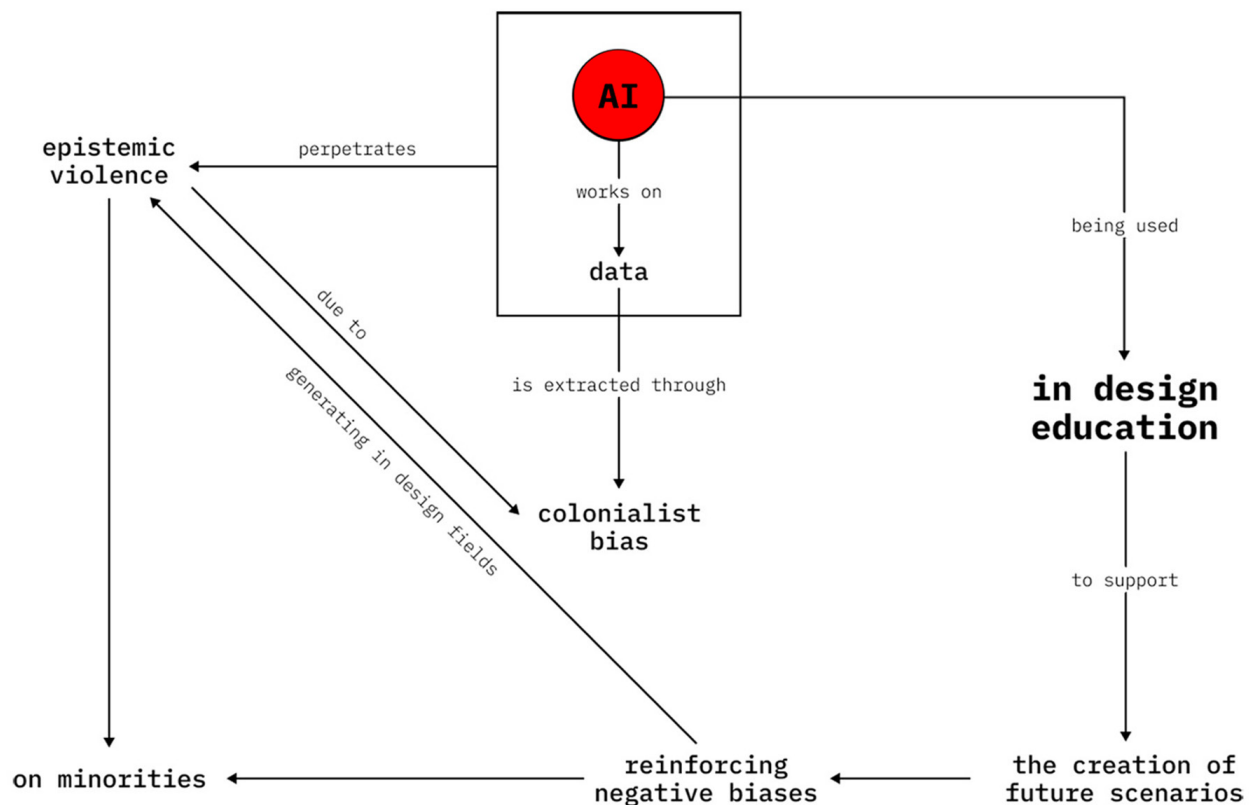
ABSTRACT In times of crisis, imagining futures becomes crucial for understanding how we want our world to develop. Design bears responsibility to support futuring practices, yet as methods evolve, examining how emerging tools can enable more decolonial approaches becomes essential.

Artificial intelligence presents opportunities for design education, increasingly used for speculation and research. However, current AI systems, built on datasets historically extracted and organized through colonialist frameworks, perpetuate epistemic violence by marginalizing non-Western perspectives. This is particularly concerning in design education, where Western programs already reproduce capitalist-colonial worldviews. Design futuring itself emerged from Western-centric traditions demonstrating limitations in integrating pluriversal perspectives, making biased AI integration especially problematic. As design recognizes the necessity of centering marginalized communities, uncritical AI integration risks entrenching Western approaches while excluding pluralistic ways of knowing, reinforcing hierarchies where dominant systems prevail and alternative futures remain unimagined.

Synthetic data has emerged as a proposed solution through debiasing processes. However, this approach doesn't resist colonialism but complies with it, merely changing how industry satisfies data needs while perpetuating value-neutral framings that obscure data's political nature.

This paper investigates how synthetic data can instead become a political tool resisting epistemic violence in design pedagogy, particularly for future scenario creation. By inverting debiasing logic to amplify rather than eliminate biases, this research proposes critical reflection on both user assumptions and AI system limitations. Through research-through-design, this work presents the design and anticipated outcomes of a pedagogical framework using an interface to support students in creating future scenarios while revealing how collective inputs encode colonial patterns.

The framework will be evaluated through a workshop demonstrating how participatory synthetic dataset creation makes epistemic violence visible through deliberate tool failure, fostering decolonial literacy and reflexive practices among design students, transforming synthetic data from technical fix into critical pedagogy tool.



Introduction

In times of crisis, the capacity to imagine futures, both positive and negative, becomes crucial for understanding how we want our world and society to develop. Design bears a responsibility to support these futuring practices through design futuring approaches, which encompass methodologies and practices that enable designers to critically explore, prototype, and materialize possible futures to view critically and shape the present (Kozubaev et al., 2020). As a strategic tool for anticipating change and shaping desirable outcomes, design futuring has become increasingly central to design education and practice. In this matter, speculative design is a particularly relevant field, because it allows designers to explore future scenario options and critically engage with alternative storylines through design artefacts. In design pedagogy in particular, speculative design can open the minds of young designers and can help them attune with otherness, including other cultures, societies and species.

In this matter, artificial intelligence (AI) technologies offer significant opportunities for design education, due to their non-human nature and the possibility for serendipitous discoveries through algorithmic hallucinations, incomplete associations, errors or misunderstandings. The use of AI is already widespread in design education, for students use it as research, brainstorming and sometimes as prototyping tool and assistant. However, these technologies also introduce profound ethical, cultural, and political challenges.

Since AI systems are built on data that is historically extracted, organized, and represented through colonialist frameworks, their application often perpetrates epistemic violence—which is “violence exerted against or through knowledge” (Galván-Álvarez, 2010)—through the definition of a hierarchy of knowledge systems that negatively impact marginalized groups through biased interpretation and representation (D’Ignazio & Klein, 2020).

AI is based on data, which is extracted, organised and represented through a colonialist bias. As a consequence, its uses generate epistemic violence against minorities. AI is frequently used in design education, both by educators and students, to support the creation of future scenarios. This reinforces negative biases against minorities in these scenarios, further perpetrating epistemic violence also in design fields.

Data colonialism can result in epistemic erasure of those excluded from the data extraction practices because of the imposition of inadequate data as knowledge, ignoring the lived experiences of those harmed, which comes embedded in the colonialist practices which are systemically excluding the South from the mainstream knowledge system by imposing production and extraction hierarchies on Indigenous communities (Brunner, 2021).

To address this matter, synthetic data has emerged as a proposed solution. Synthetic data refers to data generated algorithmically, often to statistically resemble an original dataset, but more broadly to create structured, generative data representations derived from existing inputs. It can be derived from real datasets (through augmentation processes) or be fully synthetic. Synthetic data is increasingly promoted in policy and industry discourse, including within European AI regulatory debates, as a mechanism for improving fairness and privacy, specifically as a debiasing tool. The debiasing process doesn't, however, resist data colonialism, but complies with it by only changing the ways in which the industry satisfies a need for data (Couldry & Mejias, 2019), erasing complexity in favour of fairness and simplifying the intersectionality of real people's identities even more than real data does by reducing structural inequalities to statistical parity metrics. Additionally, debiasing through synthetic datasets perpetuates the idea that data is apolitical, because it is based on a form of algorithmic fairness that describes the achievement of statistical equity (D'Ignazio & Klein, 2020). This ignores the fact that such biases reflect discrimination and bias in real life (Noble, 2018), which the extra layer of production that synthetic data carries may even enhance (Toh & Park, 2025).

This research therefore proposes using synthetic data as a political resource to resist data colonialism and expand the imaginative space of design beyond Western, capitalist-colonial horizons, through reflexive practices based on the principles of situatedness, from feminist theory and feminist data studies. Instead of using synthetic data for debiasing, it is used to highlight the 'ghosts' of the da-

taset, missing knowledge both in the inputs of the user and in the AI system.

Through research-through-design, this project presents a critical pedagogy framework for co-speculation with AI aimed at fostering decolonial literacy in speculative design. This framework makes use of an interface that uses synthetic data amplification to help students recognize knowledge hierarchies embedded in their own assumptions and in AI tools. Through deliberate tool failure, students develop critical awareness of how data colonialism shapes both their imagination and computational systems, laying groundwork for more situated and reflexive speculative practices.

This framework is developed specifically for Politecnico di Milano, a Western European technical university where the vast majority of faculty and students are Italian or European. This context is deliberately chosen: the framework's pedagogical effectiveness relies on working with students whose education has been shaped by Western epistemic traditions, making the revelation of collective Western-centric biases both pedagogically impactful and theoretically significant.

This research contributes:

- Novel use of synthetic data amplification for critical pedagogy;
- Framework demonstrating data colonialism through direct experience;
- Method for collective dataset creation in educational contexts;
- Transferable workshop methodology for design educators.

This paper proceeds as follows: Section 2 establishes research objectives and learning objectives. Section 3 details the research approach and workshop methodology. Section 4 presents anticipated results based on the future application of the framework. Section 5 concludes with implications for design education and future research directions.

Aims and objectives

Research Aim

The aim of this experiment is to investigate an approach to understand how synthetic data can be used to resist epistemic violence through a decolonial lens in a pedagogical framework dedicated to ideating future scenarios.

Research Objectives

- Develop a workshop framework that uses synthetic data to make epistemic exclusion in AI-assisted design futuring experientially visible;
- Reframe synthetic data as a political intervention rather than a neutral debiasing tool;
- Enable students to experience how Western education shapes collective datasets and imaginaries.

Learning Objectives

- Identify collective biases;
- Articulate excluded futures;
- Understand data colonialism;
- Demonstrate reflexive awareness.

Methods and procedure

Research Approach

This research adopts a research-through-design methodology (Zimmerman et al., 2007), treating the development of the pedagogical framework as both process and outcome. The iterative design of the workshop, technical system, and evaluation instruments generates knowledge about how synthetic data can function as a critical pedagogy tool.

Grounded in critical pedagogy (Freire, 1974), the framework positions learning as confrontation with systems of oppression rather than neutral knowledge transfer. Students function as co-creators, with their curated dataset serving as both object of study and learning material. Speculation becomes both the playground where epistemic violence becomes visible and problematic, and the methodology that offers the critical tools necessary to puncture this singular reality.

The study will be conducted with a small sample of students at Politecnico di Milano in May 2026.

Politecnico di Milano is a technical university in Milan, Italy, deeply rooted in Western European academic traditions, with design curricula emphasizing approaches characteristic of Western design education. This context is central to the framework's design. The pedagogical intervention specifically requires participants whose collective assumptions have been shaped by Western education. The framework is designed to surface patterns that reflect the systematic re-

production of Western epistemologies in design education.

Haraway's (1988) situated knowledges argues that all knowledge is produced from particular positions. Rather than claiming universality, this research explicitly positions itself as examining how Western design education produces Western-centric designers, with the goal of revealing the 'god trick' (Haraway, 1988) of data systems to reveal their human, and therefore fallible, nature. The god trick is revealed by making the ghosts of the dataset emerge through a hauntological approach, following Derrida's idea of hauntology (Fisher, 2014; Patil et al., 2024). Particularly, colonial hauntology deals with showing memories and presences that can't be represented but can't be laid to rest, since the events that unleashed them are not ended (Sterling, 2021). In this project, hauntology names the return of excluded futures and missing knowledge when the interface fails to generate or stabilise a scenario. The failure is not treated as a technical malfunction, but as the pedagogical condition through which absent perspectives become visible. By making the system overemphasise dominant narratives until it breaks, the workshop stages a spectral encounter with what the technological infrastructure has left out.

Positionality Statement

This research is conducted by a white Western scholar trained within European academic institutions. The critique of Western epistemic dominance therefore emerges from within, rather than outside, the system it interrogates. Following Haraway's (1988) argument that all knowledge is situated, this project does not claim a neutral or decolonized standpoint. Instead, it acknowledges its own embeddedness in Western design education and approaches synthetic data amplification as a reflexive method for confronting that complicity.

Workshop Design

This research adopts a reThe workshop is planned to follow three phases: Collective Dataset Creation, Amplification & Breakdown, and Critical Integration. This structure is designed to progressively reveal epistemic violence through collective data creation, bias amplification, and critical reflection.

Phase 1: Collective Dataset Creation

The workshop begins with theoretical framing introducing concepts of data colonialism (Coudry & Mejias, 2019), epistemic violence (Galván-Álvarez, 2010; Spivak, 1994), and synthetic data. Students are informed that their inputs will become data analyzed collectively.

After the introduction, the students, in groups, will be asked a series of questions regarding their speculative scenario. This input phase is to gather structured data from the scenarios, balancing data extraction with brainstorming to create a dataset reflecting the ideas of the students.

Following a break, the aggregated dataset is shown through a visualisation of the patterns emerged from the first phase. The students are invited to an initial discussion to reflect with prompts such as: “What patterns surprise you?”, “What’s missing from our collective dataset?”.

Phase 2: Amplification & Breakdown

A synthetic dataset is generated with patterns amplified, which will show the assumptions inserted by the students but structured and distorted to better show their shortcomings. The students will use the tool to co-speculate potential future scenarios. They are specifically tasked with trying to break the tool. Scenario generation demonstration shows how the collective dataset produces amplified scenarios. Interactive highlighting shows which dataset attributes influenced each element.

Student experimentation provides individual access to the interface via shared link. If students attempt to create scenarios including missing perspectives, the system either continues amplifying biases in their revisions or begins producing error messages. As pedagogical climax, the tool enters a failure state that make absent perspectives present through their systematic exclusion, staging a hauntopological encounter.

Phase 3: Critical Integration

A discussion inviting critical reflection is created, with prompts such as: “Why can’t the tool generate these futures?”, “How does this mirror how real AI datasets are built?”. The situated reflection is designed to facilitate collective discussion, guided by questions that connect the experience to theoretical concepts. The discussion will be prompted starting from a reflection

on how data biases are born, how they fit in a frame of data colonialism and how this perpetrates discriminatory power dynamics between different knowledge systems.

The workshop ends with a series of open-ended recommendations concerning the way students can cultivate the ability to envision a range of future scenarios, drawing from a variety of knowledge sources and adopting an alternative critical stance towards AI tools.

Data Collection

Data collection and analysis will be performed combining a reflexive thematic analysis of the participants’ exit interviews and of the field notes take on the facilitation, with multimodal artifact analysis of the artefacts produced during the workshop.

Technical Implementation

The technical system is designed to make data aggregation visible, amplify biases reliably and fail dramatically when confronted with excluded perspectives.

Conventional synthetic data debiasing operates through data augmentation: protected attributes susceptible to bias are identified, along with proxy attributes dependent on them. A synthetic dataset is then generated where fairness constraints regulate protected attribute proportions to achieve statistical balance, while original dataset proportions determine remaining attributes. Proxy attributes are subsequently evaluated for bias reduction.

This framework inverts this logic: applying bias constraints that amplify overrepresented patterns and explicitly exclude underrepresented ones. While the interface is still a work in progress at the moment of writing, the planned technical implementation involves four phases:

- Scenario coding: After the extraction template, each scenario is coded into a small set of narrative variables: setting, protagonist, institution, technology, time horizon, assumed normality, excluded perspective, and dependency;
- Pattern selection: The system identifies the most repeated or dominant combinations across the class dataset, therefore not ‘detecting bias’ in a

statistical fairness sense, but identifying dominant narrative markers;

- Amplification rules: The generator rewrites the scenario by over-representing the dominant pattern, flattening diversity where the dataset is already narrow, and surfacing the omitted elements as gaps, contradictions, or distortions;
- Sparring output: The interface shows the original scenario beside the amplified one, plus a trace of which coded variables drove the transformation. When students revise the scenario, the system either intensifies the same pattern or fails, so the dominant narrative becomes hard to ignore.

their own imaginaries. Rather than ‘correcting’ bias, the framework aims to make its structuring force visible.

For these reasons, the evaluation of the workshop will, according to Research-through-design literature, be evaluated through the following criteria:

- Decolonial literacy gained;
- Assumptions surfaced;
- Agency and reflexivity;
- Speculative range expanded;
- Critical rigor of the RtD process.

As the interface remains available as a co-speculation tool, it is expected that students may iteratively construct scenarios that are more explicitly situated and plural in epistemic reference.

Expected Results

This section outlines expected outcomes derived from the framework (Couldry & Mejias, 2019; Haraway, 1988) and existing critiques of Western design education (Escobar, 2018). The workshop implementation, scheduled for May 2026, will empirically examine these projections.

As the participants will be design students at the Politecnico di Milano, the workshop is expected to reveal patterned tendencies in students’ speculative futures. Specifically, initial scenarios may privilege techno-solutionist imaginaries, innovation-driven narratives, and market-oriented frameworks, while marginalizing or omitting intersectional perspectives and alternative economic or political models beyond capitalism.

These anticipated tendencies will not be framed as individual shortcomings, but as reflections of broader epistemic hierarchies embedded within Western design curricula (Escobar, 2018). In this sense, the workshop may surface dynamics aligned with what Couldry and Mejias (2019) describe as data colonialism and what decolonial scholars identify as epistemic exclusion, the structural absence of certain knowledge systems from dominant imaginaries.

The primary pedagogical expectation is the cultivation of reflexive awareness. Through engagement with the speculative companion, students are expected to develop the capacity to identify recurring assumptions, articulate their epistemic positioning, and recognize the limits of

Conclusions

This research presents a pedagogical framework that repositions synthetic data from a technical instrument of bias mitigation to a political and critical resource within design education. Rather than treating bias as a defect to be corrected through statistical balancing, the framework inverts conventional debiasing logics and uses amplification to make patterned exclusions visible. Through a research-through-design approach, the future workshop will analyse how collective assumptions, when formalized into datasets, can reproduce dominant epistemic structures and constrain what futures become imaginable.

Theoretically, this work advances discussions on data colonialism and situated knowledges by translating them into an experiential format within design pedagogy. By staging moments of breakdown and deliberate tool failure, the framework makes structural asymmetries in knowledge production perceptible rather than abstract. Methodologically, it introduces bias amplification as pedagogical strategy, positioning students simultaneously as data producers and critical interpreters of their own computational artifacts. Practically, it offers a transferable workshop model for educators seeking to foster reflexive AI engagement in speculative design contexts.

Importantly, this research does not claim to eliminate bias, but to cultivate the capacity to recognize how imagina-

tion is shaped by infrastructural, historical, and educational conditions.

The framework is intentionally situated within a Western European design institution, acknowledging that its effectiveness relies on engaging participants shaped by Western academic traditions. This specificity limits generalizability but strengthens the study's commitment to situated inquiry. Future research may explore adaptations in different educational contexts, and longitudinal studies

assessing whether reflexive awareness translates into sustained changes in practice. Rather than offering a solution to data colonialism, this work proposes a method for resisting it in a design education setting. By transforming synthetic data into a site of critical encounter, the framework lays groundwork for more situated, plural, and accountable approaches to design futuring, offering foundation for ongoing decolonial work.

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